Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 12: Surface Areas and Volumes (Core-Basics Edition)

CHAPTER 12: SURFACE AREAS AND VOLUMES

A simplified, detailed guide in Hinglish
Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) SURFACE AREA vs VOLUME
Surface Area = bahar ki area (paint, unit cm^2).
Volume = andar ki jagah (paani, unit cm^3).

2) CSA vs TSA
CSA = Curved/side surface (top-bottom chhod ke).
TSA = Total surface (sab milake).

3) pi (PI)
pi = 22/7 (radius 7 ka multiple) ya 3.14.

4) SQUARE AUR CUBE

a^2 = a x a (area). a^3 = a x a x a (volume).

5) SQUARE ROOT (slant height ke liye)
sqrt(r^2 + h^2). Cone mein slant height aise nikalti.

6) BASIC SHAPES KE NAAM

Cube, Cuboid, Cylinder (belan), Cone (shanku),
Sphere (gol), Hemisphere (aadha sphere).

7) UNITS

Area -> cm^2, Volume -> cm^3. Mat bhoolna.

NOW: ASLI CHAPTER 12 SHURU

TOPIC 1: FORMULA SHEET (yaad rakh)

EXAM ALERT: Saare formula ratta maar - chapter inhi pe.

CUBE (side a): TSA = 6a^2 ; Vol = a^3
CUBOID (l,b,h): TSA = 2(lb+bh+hl) ; Vol = lbh
CYLINDER (r,h): CSA = 2 pi r h ; TSA = 2 pi r(r+h)
Vol = pi r^2 h
CONE (r,h,l): CSA = pi r l ; TSA = pi r(l+r)
Vol = (1/3) pi r^2 h ; l = sqrt(r^2+h^2)
SPHERE (r): SA = 4 pi r^2 ; Vol = (4/3) pi r^3
HEMISPHERE (r): CSA = 2 pi r^2 ; TSA = 3 pi r^2
Vol = (2/3) pi r^3

TOPIC 2: COMBINATION OF SOLIDS

EXAM ALERT: Class 10 mein solids JODKE aate hain
(ice-cream = cone+hemisphere, capsule = cylinder+2 hemi).

GOLDEN RULE:

- Surface area: sirf DIKHNE wale (exposed) surface add.
- Volume: hamesha simply ADD karo.

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples poori tarah solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Combination volume

(Exam favourite - ice cream)

Cone (r=3.5, h=12) ke upar hemisphere (r=3.5). Total volume? (pi=22/7)

Cone vol = (1/3)(22/7)(3.5)^2(12)
= (1/3)(22/7)(12.25)(12) = 154 cm^3
Hemi vol = (2/3)(22/7)(3.5)^3
= (2/3)(22/7)(42.875) = 89.83 cm^3
Total = 154 + 89.83 = 243.83 cm^3 (approx).

Solved Example 2 - Combination surface area

(Exam favourite - capsule)

Capsule = cylinder (r=3.5, h=10) + 2 hemisphere (r=3.5). Total surface area? (pi=22/7)

Cylinder CSA = 2(22/7)(3.5)(10) = 220
2 hemi = 2 x 2(22/7)(3.5)^2 = 2 x 77 = 154
Total SA = 220 + 154 = 374 cm^2.

(Cylinder ke flat ends count NAHI - hemi se dhake.)

Solved Example 3 - Cone slant + volume

Cone r=6, h=8. Slant l aur volume? (pi=3.14)

l = sqrt(6^2 + 8^2) = sqrt(100) = 10
Vol = (1/3)(3.14)(36)(8) = 301.44 cm^3.

Solved Example 4 - Cylinder volume + CSA

Cylinder r=7, h=10. Vol aur CSA? (pi=22/7)

Vol = (22/7)(49)(10) = 1540 cm^3
CSA = 2(22/7)(7)(10) = 440 cm^2.

Solved Example 5 - Sphere SA + volume

Sphere r=7. SA aur volume? (pi=22/7)
  SA = 4(22/7)(49) = 616 cm^2
  Vol = (4/3)(22/7)(343) = 1437.33 cm^3 (approx).

Solved Example 6 - Cube

Cube side 5. TSA aur volume?
  TSA = 6(5)^2 = 150 cm^2
  Vol = 5^3 = 125 cm^3.

Solved Example 7 - Cuboid

Cuboid 4x3x2. TSA aur volume?
  TSA = 2(4x3 + 3x2 + 2x4) = 2(12+6+8) = 52 cm^2
  Vol = 4 x 3 x 2 = 24 cm^3.

Solved Example 8 (EASIEST) - Hemisphere volume

Hemisphere r=3. Volume? (pi=3.14)
  Vol = (2/3)(3.14)(27) = 56.52 cm^3.

Q AND A TIME

Q1. CORE: cylinder ka volume aur CSA formula likho.

Q2. Cube side 6. TSA aur volume nikaal.

Q3. Cylinder r=7, h=20. Volume. (pi=22/7)

Q4. Cone r=9, h=12. Slant height l nikaal.

Q5. Sphere r=3. Volume. (pi=3.14)

Q6. Capsule (cylinder + 2 hemisphere) ka surface area
    kaise nikaalte hain? Logic bata.

Q7. CORE CHECK: area aur volume ke units kya hote hain?

SUMMARY

1. Cube: TSA=6a^2, Vol=a^3. Cuboid: TSA=2(lb+bh+hl), Vol=lbh.

2. Cylinder: CSA=2 pi r h, Vol=pi r^2 h.

3. Cone: CSA=pi r l, Vol=(1/3)pi r^2 h, l=sqrt(r^2+h^2).

4. Sphere: SA=4 pi r^2, Vol=(4/3)pi r^3.
   Hemisphere: TSA=3 pi r^2, Vol=(2/3)pi r^3.

5. Combination: SA -> only exposed add; Volume -> add.

CORE (RED page) revise: SA vs Volume, CSA vs TSA, pi, square vs cube, slant height sqrt, units.
